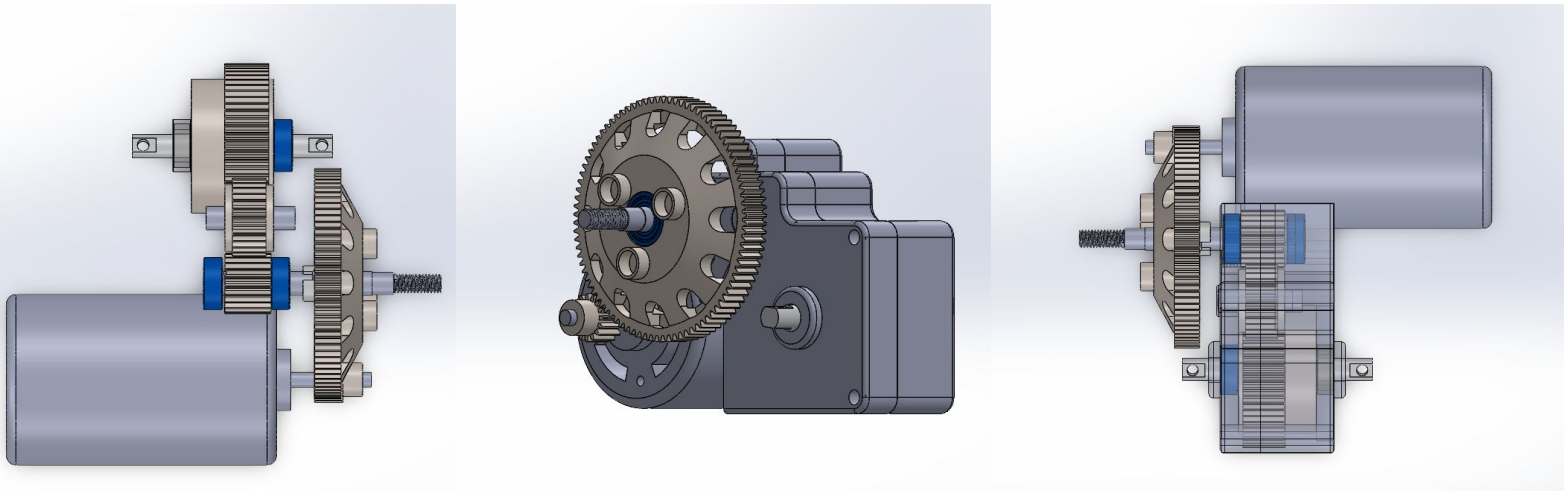


RC CAR GEARBOX/DIFFERENTIAL REDESIGN AND GEAR MODELING

- Calculated critical gear geometry for 7 gears using profile projector, calipers, MOP(Measurement Over Pins), and specialized gear calculator
- Used gear calculator values to build CAD models with accurate gear geometry
- Used a 3D model of the gear train to successfully design a gearbox housing which will be better compatible for a custom RC design project



SolidWorks Gearbox Housing Assembly



Original Gearbox Housing and Gears

SPUR GEARS - GEOMETRY						
Customer: Sample Gear Calcs		Version: 7.2		Tell us what you think !		
Description: Gear 4 and Gear 5 Mesh		Licensed To: #####		Telephone (519) 885-		
RED BOLD - INPUT DATA		email: admin@webgearservices.com				
BLUE REGULAR - CALCULATED		Drive	Driver	Tolerance		Driven
				+	-	
Gear Geometry						
# of Teeth			22			30
Module (= 25.4/ Diametral Pitch)	1.27					
Diametral Pitch (# per inch)	48					
Circular Pitch (in)		0.065				0.065
Standard Pitch Diameter (in)		0.458				0.625
Standard Center Distance (in)	0.542					
Operating Center Distance of Housing (in)	0.542		0.002	0.002		
Effective Operating Center Distance (in)	0.542		0.002	0.002		
Center Distance Extension (in)	0.000		0.002	0.002		
Operating Pitch Diameter (in)		0.458	0.002	0.002	0.625	0.002
Plot Rotation (% Rotation of Circular Pitch)	0.00%					
Pressure Angle (Deg.)	20					
Coefficient of Friction	0.08					
Base Circle Dia. (in)		0.431				0.587
Base Pitch (in)	0.062					

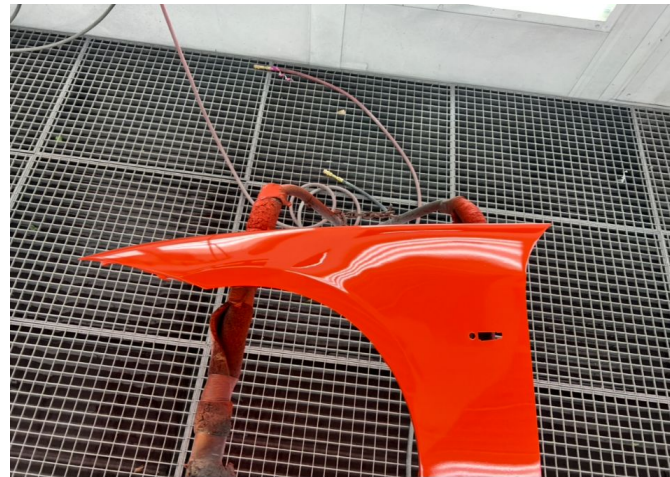
Gear Calculations

BMW E9X 3-SERIES FENDER REPAIR WITH SOLIDWORKS MODEL

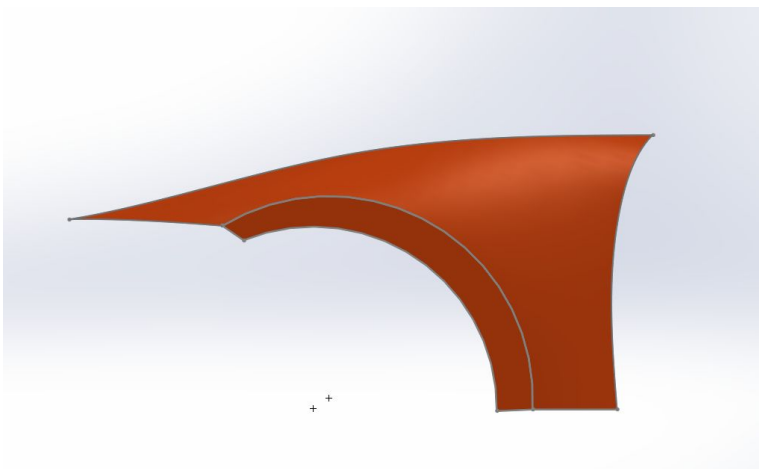
- Developed the fender model using SolidWorks surface fill, 3D-sketch, and appearance tools
- The fender took a blow from a hammer along the bodyline. The bodyline was reformed using an auto body hammer and dolly to bring back the general shape
- Continued reworking the repaired area using body filler and finishing putty to eliminate surface imperfections, then primed the panel to ensure straightness, smoothness, and corrosion protection
- Applied automotive paint and clear coat in a professional paint booth
- Used 2009 BMW 3-Series blueprints to create top, bottom, and side reference views for fender modeling



Body filler application



Final paint coat



SolidWorks Model



Final Version

FORD F-150 & 4TH GEN CAMARO FULL BODY RESTORATION AND PAINT JOB

- Worked to fully rust-proof, develop rocker panels, and perform repair work on damaged bodywork on the Ford F-150
- Primed the bedside of the truck, and sanded down for a proper finish for the final paint job
- Disassembled the Camaro for a paint job, ensuring everything could be reassembled quickly
- Prepped the front plastic bumper for paint and primed the bumper
- Sanded down the surface of the Camaro to remove previous damage
- Masked and prepped the Camaro for the final paint job



Bumper Prep



Rust Removal



Final Paint Job



Paintwork on smaller parts for F-150